

CONTACT

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PROFESSIONAL APPOINTMENTS

Postdoctoral Fellow	KICP, University of Chicago, 2024 –
Graduate Research Assistant	Johns Hopkins University, 2020–2024
Research Assistant	Peking University, 2017–2018
Visiting Research Scholar	Univ. of Michigan, Ann Arbor, Fall 2016

RESEARCH INTERESTS

physical cosmology; cosmic microwave background; CMB polarimetry; gravitational lensing; large-scale structures; statistics; high-performance computing

EDUCATION

Ph.D. Astrophysics. Johns Hopkins University. Advisor: Tobias A. Marriage	2018–2024
B.Sc. Physics. Peking University. Advisor: Joel N. Bregman	2013–2017

COLLABORATION

BICEP/Keck Collaboration	2025 –
South Pole Telescope (SPT) Collaboration	2024 –
Cosmology Large Angular Scale Surveyor (CLASS) Collaboration	2018 –

REFERENCES

Tobias A. Marriage Professor marriage@jhu.edu 2018 –	Johns Hopkins Univ.
Charles L. Bennett Bloomberg Distinguished Prof. cbennett@jhu.edu 2018 –	Johns Hopkins Univ.
Adam D. Hincks Assistant Professor adam.hincks@utoronto.ca 2020 –	University of Toronto
Renxin Xu Professor r.x.xu@pku.edu.cn 2017 –	Peking University
Joel N. Bregman H.D. Curtis Prof. jbgregman@umich.edu 2016 –	Univ. of Michigan, Ann Arbor

SELECTED AWARDS

KICP Postdoctoral Fellowship	University of Chicago, 2024
May-the-Fourth Scholarship	Peking University, 2016
CSC Visiting Fellowship	China Scholarship Council, 2016
Academic Excellence Award	Peking University, 2016
Social Work Award	Peking University, 2014
Mizuho Scholarship	The Affiliated High School of Peking University, 2013

TALKS

Largest angular scale CMB E-mode polarization measurement with CLASS	ARC Seminar, UIUC, 2025
Largest angular scale CMB E-mode polarization measurement with CLASS	mm Universe, Chicago, 2025
Quest for the large-scale CMB polarization	IHEP Seminar, IHEP/CAS, 2025
Quest for the large-scale CMB polarization	KIAA Seminar, PKU, 2025
Measurement of the Largest-scale E-mode polarization with CLASS	KIPAC Seminar, SLAC, 2025
Measurement of the Largest-scale E-mode polarization with CLASS	Gravity Group Seminar, Princeton, 2025
Polarimetry with the CLASS telescope	KICP Chalk talk, Chicago, 2024

Efficacy of the Variable-delay Polarization Modulator: Lessons Learned from CLASS

	CMB-S4 Summer Collaboration Meeting, UIUC, 2024
<i>Towards recovery of the largest-scale polarization with CLASS</i>	AAS 243 Dissertation Talk, New Orleans, 2024
<i>Accessing the largest-scale microwave polarization with CLASS</i>	Dicke Candidate Seminar, Princeton, 2023
<i>CLASS: Recent Progress and the 40 GHz Results</i>	FPD Seminar, SLAC, 2023
<i>CLASS: Recent Progress and the 40 GHz Results</i>	W&C Seminar, JHU, 2023
<i>CLASS: Recent Progress and the 40 GHz Results</i>	Cosmology Group, CCA, 2023
<i>Atmospheric Cloud Polarization</i>	Cosmo Group, Princeton, 2023
<i>Constraints on CMB Temperature evolution with SZ clusters</i>	W&C Seminar, JHU, 2020
<i>The properties of the Galactic hot gaseous halo from X-ray emission</i>	AAS 231, DC, 2018
<i>Is Calvera a strangeon star?</i>	AAS 231, DC, 2018
<i>The hot gaseous halo of the Milky Way including radiative transfer effects</i>	AAS 229, Grapevine, 2017

MEDIA

Seeing the Big Bang's Cosmic Background	NSF Podcast
Light from dawn of the universe observed by Earth-based telescopes	NSF Stories
Earth-based telescopes offer a fresh look at cosmic dawn	Johns Hopkins Hub

TEACHING

Introduction to Galaxies and Active Galactic Nuclei (TA, 171.314)	Johns Hopkins Univ., Spring, 2020
General Physics/Lab II (TA, 171.104)	Johns Hopkins Univ., Spring, 2019

OUTREACH

Guest astronomer at the Adler Planetarium (30+ hours)	Chicago, IL, 2024-
Johns Hopkins Physics Fair	Baltimore, MD, 2019
Introduction to Astronomy and Physics	Tangnan Second Middle School, Jiangxi, China, 2014

PUBLICATIONS

h-index: 14, citations: 623

AS MAIN AUTHOR

9. A Measurement of the Largest-scale CMB E-mode Polarization with CLASS
ApJ 986, 111 (2025) | YL & CLASS Collaboration
8. CLASS Angular Power Spectra and Map-component Analysis for 40 GHz Observations through 2022
ApJ 963, 92 (2024) | Eimer, Joseph R.; YL & CLASS Collaboration
7. CLASS Observations of Atmospheric Cloud Polarization at millimeter Wavelengths
ApJ 958, 154 (2023) | YL & CLASS Collaboration
6. CLASS Data Pipeline and Maps for 40 GHz Observations through 2022
ApJ 956, 77 (2023) | YL & CLASS Collaboration
5. Constraining Cosmic Microwave Background Temperature Evolution With Sunyaev-Zel'dovich Galaxy Clusters from the Atacama Cosmology Telescope
ApJ 922, 136 (2021) | YL; Hincks, Adam D. & ACT Collaboration
4. Two-year Cosmology Large Angular Scale Surveyor (CLASS) Observations: 40 GHz Telescope Pointing, Beam Profile, Window Function, and Polarization Performance
ApJ 891, 134 (2020) | Xu, Zhilei; Brewer, Michael K.; Fluxá Rojas, Pedro; YL & CLASS Collaboration
3. Two-year Cosmology Large Angular Scale Surveyor (CLASS) Observations: A Measurement of Circular Polarization at 40 GHz
ApJ 889, 105 (2020) | Padilla, Ivan L.; Eimer, Joseph R.; YL & CLASS Collaboration
2. Calvera: A Low-mass Strangeon Star Torqued by Debris Disk?
ApJ 854, 165 (2018) | YL; Wang, Weiyang; Ge, Mingyu; Liu, Xiongwei; Tong, Hao & Xu, Renxin
1. The Properties of the Galactic Hot Gaseous Halo from X-Ray Emission
ApJ 849, 105 (2017) | YL & Bregman, Joel

AS CONTRIBUTING COAUTHOR

20. Measurement of the Full Shape of the Thermal Sunyaev-Zel'dovich Power Spectrum from the South Pole Telescope and Herschel/SPIRE Observations
ApJL 1002, L22 (2026) | Raghunathan, S. & **SPTpol/SPT-3G Collaboration**
19. SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field
arXiv 2603.20163 (2026) | Quan, W.; Camphuis, E.; Daley, C.; Huang, N.; Omori, Y.; Guidi, F.; Anderes, E. & **SPT-3G Collaboration**
18. SPT-3G D1: Compton- y maps using data from the SPT-3G and Planck surveys
arXiv 2602.11279 (2026) | Maniyar, A. S.; Bianchini, F.; Wu, W. L. K.; Raghunathan, S. & **SPT-3G Collaboration**
17. Characterization of the Polarization Beam Response of SPT-3G Using Point Sources
arXiv 2602.06334 (2026) | de Haan, T.; Archipley, M.; Huang, N. & **SPT-3G Collaboration**
16. SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power
arXiv 2601.20551 (2026) | Chaubal, P.; Huang, N.; Reichardt, C. L. & **SPT-3G Collaboration**
15. SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data
PhRvD 112, 123520 (2025) | Zebrowski, J. A.; Reichardt, C. L. & **SPT-3G Collaboration**
14. Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky
arXiv 2510.24669 (2025) | Vitrier, A.; Fichman, K.; Balkenhol, L.; Camphuis, E.; Guidi, F. & **SPT-3G Collaboration**
13. PROTOCALC, a W-band Polarized Calibrator for Cosmic Microwave Background Telescopes: Application to Simons Observatory and CLASS
ApJS 279, 30 (2025) | Coppi, Gabriele; Dachlythra, Nadia; Nati, Federico; Dünner-Planella, Rolando; Adler, Alexandre E.; Errard, Josquin; Galitzki, Nicholas; YL; Petroff, Matthew A.; Simon, Sara M.; Sang, Ema Tsang King; Aguilar, Amalia Villarrubia; Wollack, Edward J. & Zannoni, Mario
12. High-efficiency and Low-noise Detectors for the Upgraded CLASS 90 GHz Focal Plane
ApJS 278, 56 (2025) | Núñez, Carolina & **CLASS Collaboration**
11. Sensitivity-improved Polarization Maps at 40 GHz with CLASS and WMAP Data
ApJ 971, 41 (2024) | Shi, Rui & **CLASS Collaboration**
10. HoverCal + PoloCalC: precise on-flight metrology performance results over CLASS telescope
SPIE 1310217 (2024) | Dünner, Rolando; Astori, Federico; Coppi, Gabriele; Eimer, Joseph; Errard, Josquin; Fluxá, Juan; YL; Mezzanzanica, Noemi; Nati, Federico; Petroff, Matthew; Puddu, Roberto; Rojas, Matías & Zannoni, Mario
9. Cosmology Large Angular Scale Surveyor (CLASS): 90 GHz Telescope Pointing, Beam Profile, Window Function, and Polarization Performance
ApJS 273, 26 (2024) | Datta, Rahul; Brewer, Michael K.; Denes Couto, Jullianna; Eimer, Joseph; YL & **CLASS Collaboration**
8. Microwave Observations of Venus with CLASS
PSJ 4, 154 (2023) | Dahal, Sumit & **CLASS Collaboration**
7. Testing Cosmic Microwave Background Anomalies in E-mode Polarization with Current and Future Data
ApJ 945, 79 (2023) | Shi, Rui & **CLASS Collaboration**
6. Calibration of Transition-edge Sensor (TES) Bolometer Arrays with Application to CLASS
ApJS 262, 52 (2022) | Appel, John W. & **CLASS Collaboration**
5. Four-year Cosmology Large Angular Scale Surveyor (CLASS) Observations: On-sky Receiver Performance at 40, 90, 150, and 220 GHz Frequency Bands
ApJ 926, 33 (2022) | Dahal, Sumit & **CLASS Collaboration**
4. Two Year Cosmology Large Angular Scale Surveyor (CLASS) Observations: Long Timescale Stability Achieved with a Front-end Variable-delay Polarization Modulator at 40 GHz
ApJ 922, 212 (2021) | Harrington, Kathleen & **CLASS Collaboration**
3. The optical/UV excess of X-ray-dim isolated neutron star II. Nonuniformity of plasma on a strangeon star surface
RAA 18, 082 (2018) | Wang, Wei-Yang; Feng, Yi; Lai, Xiao-Yu; YL; Lu, Ji-Guang; Chen, Xuelei & Xu, Ren-Xin
2. The Extended Distribution of Baryons around Galaxies
ApJ 862, 3 (2018) | Bregman, Joel N.; Anderson, Michael E.; Miller, Matthew J.; Hodges-Kluck, Edmund; Dai, Xinyu; Li, Jiang-Tao; YL & Qu, Zhijie
1. Merging strangeon stars
RAA 18, 024 (2018) | Lai, Xiao-Yu; Yu, Yun-Wei; Zhou, En-Ping; YL & Xu, Ren-Xin